

Week 3. Homework 1. SQL Basic

Due date: **September 12, 11:59 pm**

Submission Guidelines

- **Dataset:** You must use the provided *IMDB.sqlite* dataset to answer all questions.
- **Code and Output:** Copy your code and attach screenshots of your outputs under each question.
- **Submission Format:** Each student is required to push one Word or PDF file to a designated GitHub repository you create specifically for this week's assignment.
- **Repository Name:** Use the format *Week 3-HW1-LastName*, where *LastName* is your last name.
- **Code Quality:** Ensure your code is organized for easy understanding.
- **AI Usage:** You should make a serious attempt at each problem prior to using an LLM. Remember, you may be tested on SQL knowledge without assistance in a technical interview. If prompting an LLM, I suggest telling it your starting point, your thought process, and asking it to guide you. I've done this learning SQL and it is helpful and I learned more that way. Disclose your usage at the end of the file; you will not be penalized regardless, but this give me some information.

Questions

1. Directors and Their Departments

List the names of all directors along with their respective departments.

Code:

```
SELECT name, department
```

```
FROM directors
```

	name	department
1	James Cameron	Directing
2	Gore Verbinski	Directing
3	Sam Mendes	Directing
4	Christopher Nolan	Directing
5	Andrew Stanton	Directing
6	Sam Raimi	Directing
7	Byron Howard	Directing

```
Execution finished without errors.
Result: 2349 rows returned in 21ms
At line 1:
SELECT name, department
FROM directors
```

2. Popular Movies

Find the titles of all movies with a popularity score greater than 5.

Code:

```
SELECT title
```

```
FROM movies
```

```
WHERE popularity > 5
```

	title
1	Avatar
2	Pirates of the Caribbean: At World's...
3	Spectre
4	The Dark Knight Rises
5	John Carter
6	Spider-Man 3
7	Tangled

Execution finished without errors.
Result: 3376 rows returned in 43ms
At line 1:
SELECT title
FROM movies

3. Directors in the Directing Department

Retrieve the names and IDs of directors who are in the 'Directing' department and have a gender value of '2'.

Code:

```
SELECT name, id
```

```
FROM directors
```

```
WHERE department = 'Directing'
```

```
AND gender = 2;
```

	name	id
1	James Cameron	4762
2	Gore Verbinski	4763
3	Sam Mendes	4764
4	Christopher Nolan	4765
5	Andrew Stanton	4766
6	Sam Raimi	4767
7	Byron Howard	4768

```

Execution finished without errors.
Result: 1574 rows returned in 6ms
At line 1:
SELECT name, id
FROM directors

```

4. Directors with 'Sam' in Their Name

Find all directors whose names include the word 'Sam'.

Code:

```
SELECT name
```

```
FROM directors
```

```
WHERE name LIKE '%Sam%'
```

	name
1	Sam Mendes
2	Sam Raimi
3	Sam Fell
4	Sam Weisman
5	Karyn Kusama
6	Sam Taylor-Johnson
7	Sam Miller

```

Execution finished without errors.
Result: 11 rows returned in 4ms
At line 1:
SELECT name
FROM directors

```

5. First 10 Movies by Release Date

Show the first 10 movies in the database, sorted by release date in ascending order.

Code:

```
SELECT title
FROM movies
ORDER BY release_date ASC
LIMIT 10;
```

	title
1	Intolerance
2	The Big Parade
3	Metropolis
4	Pandora's Box
5	The Broadway Melody
6	Hell's Angels
7	A Farewell to Arms

Execution finished without errors.
Result: 10 rows returned in 6ms
At line 1:
SELECT title
FROM movies

6. 'Star' Movies with High Ratings

List all movies with the word 'Star' in the title and a vote average greater than 7.

```
SELECT title
FROM movies
WHERE title LIKE '%Star%'
      AND vote_average > 7
```

	title
1	Star Trek Into Darkness
2	Star Trek
3	Star Wars: Episode III - Revenge of ...
4	Stardust
5	Star Trek II: The Wrath of Khan
6	The Fault in Our Stars
7	Star Wars

```

Execution finished without errors.
Result: 9 rows returned in 5ms
At line 1:
SELECT title
FROM movies

```

7. Movies with Limited Revenue (1995–2000)

Find the titles of movies released between January 1, 1995, and December 31, 2000, with revenues not exceeding 10 million.

Code:

```
SELECT title
```

```
FROM movies
```

```
WHERE release_date BETWEEN '1995-01-01' AND '2000-12-31'
```

```
AND revenue <= 10000000
```

	title
1	Volcano
2	Déjà Vu
3	Mighty Joe Young
4	Little Nicky
5	Inspector Gadget
6	Anna and the King
7	Primary Colors

```

Execution finished without errors.
Result: 328 rows returned in 8ms
At line 1:
SELECT title
FROM movies

```

8. Directors with Specific 'uid' Values

List the names of directors who have a uid greater than 7,000 or less than 100.

Code:

```
SELECT name
FROM directors
WHERE uid > 7000
      OR uid < 100;
```

	name
1	Sam Mendes
2	Andrew Stanton
3	Sam Raimi
4	Byron Howard
5	Joss Whedon
6	David Yates
7	Zack Snyder

Execution finished without errors.
Result: 2071 rows returned in 7ms
At line 1:
SELECT name
FROM directors

9. Movies with Specific Vote Counts

Retrieve the titles of movies that have:

- A vote counts between 50 and 100, inclusive.
- A vote average of exactly 7.0.

Code:

```
SELECT title
FROM movies
WHERE vote_count BETWEEN 50 AND 100
      AND vote_average = 7.0;
```

	title
1	Angela's Ashes
2	A Better Life
3	Red Riding: In the Year of Our Lord ...
4	Quo Vadis
5	Another Year
6	Salvador
7	The Black Stallion

Execution finished without errors.
Result: 9 rows returned in 6ms
At line 1:
SELECT title
FROM movies

10. Directors Named 'John' with Specific 'uid' Values

Find the names of directors who:

- Have 'John' in their name.
- Have a uid between 5,000 and 10,000 (exclusive).

Code:

```
SELECT name
```

```
FROM directors
```

```
WHERE name LIKE '%John%'
```

```
AND uid > 5000
```

```
AND uid < 10000;
```

	name
1	John Lasseter
2	John Milius
3	John Singleton
4	John Guillermin
5	John McNaughton
6	John Ottman
7	John Huston

```
Execution finished without errors.  
Result: 8 rows returned in 5ms  
At line 1:  
SELECT name  
FROM directors
```

11. Movies with Odd 'uid' and Even Vote Count

Retrieve the titles of movies that meet the following criteria:

- The uid is an odd number.
- The vote count is an even number and greater than 500.

Code:

```
SELECT title
```

```
FROM movies
```

```
WHERE uid % 2 = 1
```

```
      AND vote_count % 2 = 0
```

```
      AND vote_count > 500;
```

	title
1	Avatar
2	Pirates of the Caribbean: At World's...
3	Spectre
4	John Carter
5	Spider-Man 3
6	Tangled
7	Pirates of the Caribbean: On Strange...

Execution finished without errors.
Result: 405 rows returned in 8ms
At line 1:
SELECT title
FROM movies

12. Movies with Specific Ratings or Budgets

Find the titles of movies that meet any of the following criteria:

- Have a vote average between 6.0 and 8.0, with a vote count of exactly 200.
- OR have a budget greater than 100,000 but less than 500,000.

Code:

SELECT title

FROM movies

WHERE (vote_average BETWEEN 6.0 AND 8.0

AND vote_count = 200)

OR (budget > 100000

AND budget < 500000);

	title
1	Halloween
2	Control
3	Snow Flower and the Secret Fan
4	Night of the Living Dead
5	Sleeper
6	Open Water
7	Conversations with Other Women

```
Execution finished without errors.  
Result: 58 rows returned in 13ms  
At line 1:  
SELECT title  
FROM movies
```